Speak **L**ike **A** **M**athematician:

corresponding angles:

alternate interior angles:

alternate exterior angles:

consecutive interior angles:

Corresponding Angles Postulate

If two ___________________ lines are cut by a transversal, then the pairs of corresponding angles are ___________________.

Theorems — Parallel Lines and a Transversal

Alternate Interior Angles Theorem: the pairs of alternate interior angles are ___________________.

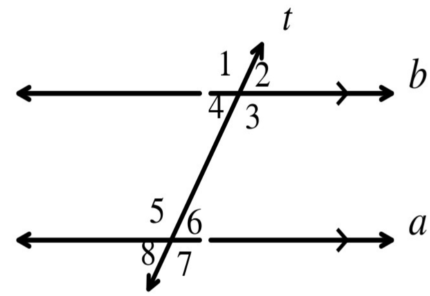
Alternate Exterior Angles Theorem: the pairs of alternate exterior angles are ___________________.

Consecutive Interior Angles Theorem: the pairs of consecutive interior angles are ___________________.

SC Honors Geometry - Nappier
3.2 Parallel Lines and Transversals

Example 1 – Using Angle Relationships

In the diagram, a ∥ b. Find each angle measure and name the relationship you used.



a) m∠1 = 106°. Find m∠5.

b) m∠3 = 71°. Find m∠5.

c) m∠4 = 118°. Find m∠5.

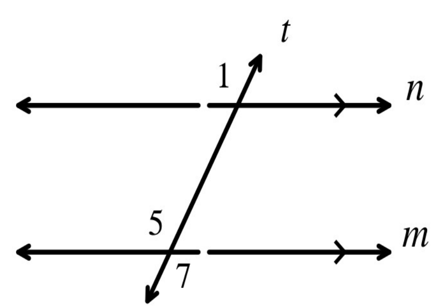
d) m∠2 = 64°. Find m∠8.

Example 2 – Proving the Alternate Exterior Angles Theorem

Given: m ∥ n

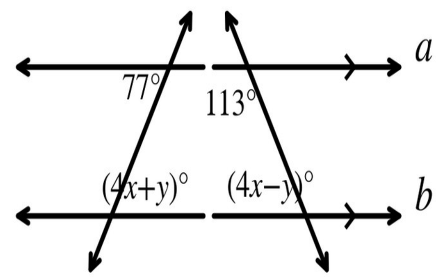
Prove: ∠1 ≅ ∠7

Statements	**Reasons**
1. m ∥ n	**1.** Given
2. ∠1 ≅ ∠5	**2.**
3.	**3.**
4.	**4.**



Example 3 – Find x and y

In the diagram, a ∥ b. Find the values of x and y. Show your work.



a) Write an equation using the 77° angle.

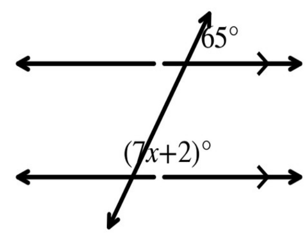
b) Write an equation using the 113° angle.

c) Solve the system for x and y.

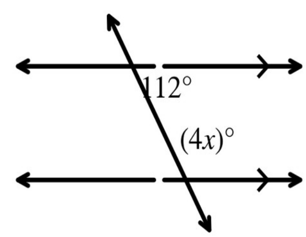
Example 4 – Find the Value of x

Each pair of lines is parallel. Find x and name the relationship you used.

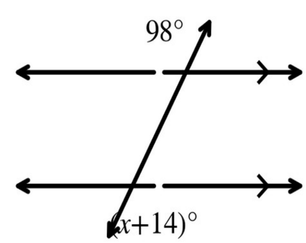
a)



b)

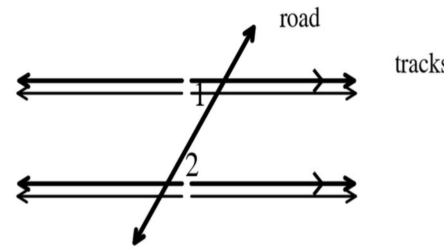


c)



Example 5 – A Real-Life Application

A straight road crosses two parallel railroad tracks. The road makes a 41° angle with the first track.



a) Copy the diagram and label m∠1 = 41°. Which angle pair do ∠1 and ∠2 form?

b) Find m∠2. Justify your answer.

c) Find the measure of the angle that is consecutive interior with ∠1.

Example 6 – Explain the Error

A classmate is given the diagram below with no parallel marks.

a) The classmate says "∠3 ≅ ∠7 because corresponding angles are congruent." What is wrong?

b) What information would make the classmate correct?